STA2002 Probability & Statistics II Lecture 3 - 2024-2025 Term 1

Lecture 3: Confidence Interval

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Suggested Reading: Hogg et al. 2018, Chapter 7.

Point Estimation vs. Interval Estimation: In the previous lecture we learned how a parameter can be estimated from sample data. However, it is also important to understand how good is the estimate obtained, since the estimate is a random quantity due to sampling variability. For example, suppose we want to estimate the mean of the normal distribution μ using sample mean $\bar{x} := \frac{1}{n}\sum_i x_i$, however, it is almost never the case that $\mu = \bar{x}$ (Why?), and the point estimate $\bar{x}$ says nothing about how close $\bar{x}$ is to μ. When $\bar{x} = 100$, is the mean μ likely to between 90 and 110? Or is it likely to between 80 and 120? Such bounds that represent an interval of plausible values for a parameter are an example of an **interval estimate**.

An interval estimate for a population parameter is called a **confidence interval**. Meanwhile, We cannot guarantee that the interval contains the true, unknown population parameter — because we only use a sample from the full population to compute the point estimate and the interval. However, the confidence interval is constructed so that we have high confidence that it does contain the unknown population parameter.

In order to derive the confidence interval, the sampling distribution of the estimator is usually used. Therefore we first recall the following results: Recall the sampling distribution

Theorem 0.1 (Sampling distribution). *Suppose $X_1, \cdots, X_n$ are independent and identically distributed normal random variables with $\mathbf{E}[X_1] = \mu$ and $\mathrm{Var}(X_1) = \mathbf{E}(X_1 - \mu)^2 = \sigma^2$. Define the sample mean $\bar{X} = \frac{1}{n}\sum_{i=1}^n X_i$ and the sample variance $S^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X})^2$. Then:*

(i) $\bar{X}$ and S^2 are independent;

(ii) The distribution of $\frac{(n-1)S^2}{\sigma^2}$ is χ^2_{n-1}, the χ^2 distribution with degree of freedom $n-1$;

(iii) Thus we have $T := \frac{\sqrt{n}(\bar{X}-\mu)}{S}$ satisfies the t distribution with degree of freedom $n-1$.

To illustrate the basic ideas of confidence interval (CI), we will start with the most easily understood example of estimating the mean of a normal population, which is detailed in the first subsection below.

1 General Properties Regarding the Confidence Interval

Suppose $X_1, \cdots, X_n$ are data from a population, and θ is the parameter to be estimated. Define the confidence interval as follows:

Definition 1.1 (Confidence interval)**.** *Suppose two estimators (i.e., a function of data)* $\widehat{L}, \widehat{U}$, $\widehat{L} \leq \widehat{U}$ *satisfies*

$$\mathbf{P}_\theta \left(\widehat{L} \leq \theta \leq \widehat{U} \right) \geq 1 - \alpha \text{ for all } \theta \in \Theta$$

then we call the interval $[\widehat{L}, \widehat{U}]$ *the* $1 - \alpha$ *confidence interval for the parameter* θ*.*

This means that **there is a probability of** $1 - \alpha$ **of selecting a sample for which the resulted CI** $[\widehat{L}, \widehat{U}]$ **will contain the true value** θ. The end-points $\widehat{L}$ and $\widehat{U}$ are called the **lower- and upper-confidence limits**, and $1 - \alpha$ is called the **confidence level**.

Remark A counter-intuitive thing here is that $\widehat{L}$, $\widehat{U}$ are estimators while θ is a fixed (even though unknown) number (see figure 7.1-1 in textbook Hogg et al. 2018, also we will do numerical experiments regarding this). Then we will introduce a concept, called 'pivotal statistics', which is defined as follow:

Definition 1.2 (Pivotal quantity or Pivot)**.** *A pivot is a* ***function*** $\widehat{T}$ ***of the observations*** $X_1, \cdots, X_n$ ***and the unobservable parameter*** θ *satisfying:*

$$\widehat{T} = h(X_1, \cdots, X_n, \theta) \to_d T_0 \text{ and } T_0\text{'s distribution does not contain any unknown parameters.} \quad (1)$$

Notably, here $\widehat{T}$ is allowed to have unknown parameters. Therefore, a pivot is not necessarily a 'statistics/estimator'.

Example 1 Suppose $X_1, \cdots, X_n$ are independent normally distributed random variables with unknown mean θ and unknown variance σ^2. Define $\widehat{\theta} = \frac{1}{n} \sum_{i=1}^n X_i$ and $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \widehat{\theta})^2$. Then we have

$$\sqrt{n}(\widehat{\theta} - \theta)/\sigma \sim N(0, 1)$$

and

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$$

Therefore we have

$$\widehat{T} = \frac{\sqrt{n}(\widehat{\theta} - \theta)}{S} \sim t_{n-1}$$

which does not depend on the unknown parameters θ, σ^2, so $\widehat{T}$ is a pivot.

Similarly, if we calculate the maximum likelihood estimate $\widehat{\sigma} = \sqrt{\frac{1}{n} \sum_{i=1}^n (X_i - \widehat{\theta})^2}$, from slutsky's theorem, we have

$$\widehat{T}' = \sqrt{n}(\widehat{\theta} - \theta)/\widehat{\sigma} \to_d N(0, 1), \quad \text{as } n \to \infty$$

Because $\widehat{T}'$ only contains the data and the unknown parameter θ, so $\widehat{T}'$ is also a pivot.

Remark [**A general method for establishing a** $1 - \alpha$ **confidence interval**] Suppose $q_{\alpha/2}$ and

$q_{1-\alpha/2}$ are the **upper** $\alpha/2$ and the $1-\alpha/2$ quantile of the T_0, i.e.,

$$\mathbf{P}(T_0 \geq q_{\alpha/2}) = \alpha/2 \text{ and } \mathbf{P}(T_0 \leq q_{1-\alpha/2}) = \alpha/2$$

Suppose the cdf of T_0 is continuous, then $\mathbf{P}(T_0 < q_{1-\alpha/2}) = \mathbf{P}(T_0 \leq q_{1-\alpha/2}) = \alpha/2$. Define the confidence interval

$$\left\{u : q_{1-\alpha/2} \leq h(X_1, \cdots, X_n, \theta) \leq q_{\alpha/2}\right\}$$

Then from our lecture 1,

$$\lim_{n\to\infty} \mathbf{P}(q_{1-\alpha/2} \leq h(X_1, \cdots, X_n, \theta) \leq q_{\alpha/2}) = \mathbf{P}(q_{1-\alpha/2} \leq T_0 \leq q_{\alpha/2}) = 1-\alpha \tag{2}$$

The confidence interval is obtained by solving the above inequalities:

$$q_{1-\alpha/2} \leq h(X_1, \cdots, X_n, \theta) \leq q_{\alpha/2} \quad \Rightarrow \quad \hat{L}(X_1, \ldots, X_n) \leq \theta \leq \hat{U}(X_1, \ldots, X_n).$$

2 Confidence Intervals for the Normal Model

2.1 Variance known

We start with the easiest case where the normal variance σ^2 is perfectly known.

Suppose that $X_1, \ldots, X_n$ are iid samples from $N(\mu, \sigma^2)$, where μ is unknown (to be estimated), and σ^2 is known. Therefore, we know that

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \sim N(\mu, \frac{\sigma^2}{n})$$

And thus the **standardized** random variable (by subtracting the mean and dividing by the standard deviation):

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1) \tag{3}$$

follows the standard normal distribution. Therefore, Z is a **pivot**.

From the pivot Z we can derive the confidence interval for μ as follows. Recall from Definition 1.1 that the **two-sided confidence interval** for μ is a interval of the form $\widehat{L} \leq \mu \leq \widehat{U}$, where the end-points $\widehat{L}$ and $\widehat{U}$ are computed from the sample data (thus $\widehat{L}$ and $\widehat{U}$ are random). For given $\alpha \in (0,1)$, $\widehat{L} \leq \mu \leq \widehat{U}$ is a $1-\alpha$ confidence interval if the values $\widehat{L}$ and $\widehat{U}$ satisfy:

$$\mathbf{P}(\widehat{L} \leq \mu \leq \widehat{U}) = 1-\alpha$$

(note that here the probability is taken with respect to the randomness in the samples, $\widehat{L}, \widehat{U}$ are functions of the sample data, and μ is a fixed but unknown constant).

For the estimation of μ with known σ^2, equation (3) leads to

$$\mathbf{P}(-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}) = 1 - \alpha$$

which translates to

$$\mathbf{P}(\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}) = 1 - \alpha$$

where $z_{\alpha/2}$ is the upper $\alpha/2$ quantile of the standard normal distribution, i.e., $\mathbf{P}(Z \geq z_{\alpha/2}) = \alpha/2$. Check the z-table for the $z_{\alpha/2}$ values.

Therefore, by letting

$$\widehat{L} = \bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \quad \widehat{U} = \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

we have the CI: $\bar{X} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$ is a $1 - \alpha$ CI for μ.

Remark: The CI is not unique. For example, we may let $\widehat{L} = \bar{X} - z_{\alpha/4}\frac{\sigma}{\sqrt{n}}$ and $\widehat{U} = \bar{X} + z_{3\alpha/4}\frac{\sigma}{\sqrt{n}}$ and it is still a $1 - \alpha$ CI. Which one is better?

Interpretation: Note that in a single trial, i.e., given a sample, we calculate the CI $[\widehat{L}, \widehat{U}]$ as $\bar{X} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$. Then we have either μ is included in $[\widehat{L}, \widehat{U}]$ or μ is not included in it (μ is a constant value, not a RV). Thus it is wrong to interpret that μ is included in $[\widehat{L}, \widehat{U}]$ with probability $1 - \alpha$. The correct interpretation of a $(1 - \alpha)$ CI depends on the relative frequency view of probability. Suppose an infinite number of random samples are collected and a $1 - \alpha$ CI for μ is computed for **each sample**, then we have an infinite number of CIs, among which $1 - \alpha$ **of these intervals will contain the true value** μ.

Confidence Level and Precision of Estimation. The length of CI is $2z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$. Generally, for a fixed sample size n and standard deviation σ, the higher the confidence level $(1 - \alpha)$, the longer the resulting CI. The length of a confidence interval is a measure of the **precision of estimation**. It is desirable to obtain a confidence interval that is *short* enough for estimating purposes but also has adequate confidence. One way to achieve this is by choosing the sample size n to be large enough to give a CI of specified length or precision with prescribed confidence. In using $\bar{X}$ to estimate μ, the **error** $E = |\bar{X} - \mu|$ is at most $z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$ with probability (confidence) $1 - \alpha$. In situations where the sample size can be controlled, we can choose n such that we are $1 - \alpha$ confident that the error in estimating μ is less than a pre-specified bound E on the error. The corresponding sample size is found by choosing n such that $z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq E$, which leads to

$$n = \lceil \Big(\frac{z_{\alpha/2}\sigma}{E}\Big)^2 \rceil \tag{4}$$

where $\lceil t \rceil$ is the ceiling function that returns the smallest integer which is greater than or equal to t.

Remark: Generally, from equation (4), we have the relationship between sample size n, confidence level $1-\alpha$, length of CI $2E$, and std σ:

- For fixed σ and $1-\alpha$, the required sample size n increases as the desired error E decreases.
- For fixed E and $1-\alpha$, the required sample size n increases as the std σ increases.
- For fixed E and σ, the required sample size n increases as the level $1-\alpha$ increases.

One-sided CI: When setting either $\widehat{L} = -\infty$ or $\widehat{U} = +\infty$, we have the **one-sided** Confidence interval/Confidence bounds:

1. $\widehat{U}$ is an $1-\alpha$ **upper confidence bound** for μ if

$$\mathbf{P}(\mu \leq \widehat{U}) = 1-\alpha$$

2. $\widehat{L}$ is an $1-\alpha$ **lower confidence bound** for μ if

$$\mathbf{P}(\mu \geq \widehat{L}) = 1-\alpha$$

For the estimation of μ with known σ^2, we have

1. $\bar{X} + z_\alpha \frac{\sigma}{\sqrt{n}}$ is an $1-\alpha$ **upper confidence bound** for μ
2. $\bar{X} - z_\alpha \frac{\sigma}{\sqrt{n}}$ is an $1-\alpha$ **lower confidence bound** for μ.

2.2 Variance unknown

When estimating the normal mean μ but σ^2 is unknown, we note that

$$T := \frac{(\bar{X}-\mu)}{S/\sqrt{n}} \sim t_{n-1}$$

Therefore T is a pivot and we have

$$\mathbf{P}(-t_{\alpha/2,n-1} \leq \frac{\bar{X}-\mu}{S/\sqrt{n}} \leq t_{\alpha/2,n-1}) = 1-\alpha$$

which translates to

$$\mathbf{P}(\bar{X} - t_{\alpha/2,n-1}\frac{S}{\sqrt{n}} \leq \mu \leq \bar{X} + t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}) = 1-\alpha$$

where $t_{\alpha/2,n-1}$ is the upper $\alpha/2$ quantile of the t distribution with degree of freedom $n-1$, i.e., $\mathbf{P}(T \geq t_{\alpha/2,n-1} = \alpha/2$. Check the t-table for the $t_{\alpha/2,\nu}$ values.

To summarize:

1. $\bar{X} \pm t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}$ is the two-sided $1-\alpha$ CI for μ

2. $\bar{X} + t_{\alpha,n-1}\frac{S}{\sqrt{n}}$ is the $1-\alpha$ upper confidence bound for μ

3. $\bar{X} - t_{\alpha,n-1}\frac{S}{\sqrt{n}}$ is the $1-\alpha$ lower confidence bound for μ

Note that $t_{\alpha,\nu} \to t_\alpha$ as $\nu \to \infty$, thus in practice when ν is large, we can also use z_α instead.

2.3 Confidence Interval for Variance

Sometimes confidence intervals on the population variance or standard deviation are desired. We first consider the interval estimate of σ^2 for normal distribution.

Let $X_1, \ldots, X_n$ be iid sample from $N(\mu, \sigma^2)$, recall that

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$$

where $S^2 := \frac{1}{n-1}\sum(X_i - \bar{X})^2$ is the sample variance. Denote $\chi^2_\alpha(n-1)$ as the upper α quantile of the $\chi^2(n-1)$ distribution (check the χ^2 table for values). Then we have

$$\mathbf{P}(\chi^2_{1-\alpha/2}(n-1) \leq \frac{(n-1)S^2}{\sigma^2} \leq \chi^2_{\alpha/2}(n-1)) = 1-\alpha$$

and thus the two-sided CI for σ^2 is

$$\mathbf{P}(\frac{(n-1)S^2}{\chi^2_{\alpha/2}(n-1)} \leq \sigma^2 \leq \frac{(n-1)S^2}{\chi^2_{1-\alpha/2}(n-1)}) = 1-\alpha$$

To summarize:

1. $[\frac{(n-1)S^2}{\chi^2_{\alpha/2}(n-1)}, \frac{(n-1)S^2}{\chi^2_{1-\alpha/2}(n-1)}]$ is the two-sided $1-\alpha$ CI for σ^2

2. $\frac{(n-1)S^2}{\chi^2_{1-\alpha}(n-1)}$ is the $1-\alpha$ upper confidence bound for σ^2

3. $\frac{(n-1)S^2}{\chi^2_{\alpha}(n-1)}$ is the $1-\alpha$ lower confidence bound for σ^2

3 Asymptotic Confidence Intervals

3.1 Confidence Interval for Mean of Non-Normal Distributions

So far we have assumed that the population distribution is normal. We now present a **large-sample CI** for μ for non-normal distributions. Let $X_1, \ldots, X_n$ be a random sample from a population with unknown mean μ and variance σ^2. Now if the sample size n is large, by the CLT,

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim_{\text{approx}} N(0,1)$$

Therefore when n is sufficiently large (say $n \geq 30$), we have the **approximate** CI for μ as follows:

- When σ^2 is known, we have
  1. $\bar{X} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$ is an approximate $1-\alpha$ CI for μ
  2. $\bar{X} + z_{\alpha}\frac{\sigma}{\sqrt{n}}$ is an approximate $1-\alpha$ upper confidence bound for μ
  3. $\bar{X} - z_{\alpha}\frac{\sigma}{\sqrt{n}}$ is an approximate $1-\alpha$ lower confidence bound for μ.
- When σ^2 is unknown, we can either replace σ by the sample standard deviation S in above approximate CIs, or use
  1. $\bar{X} \pm t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}$ is an approximate $1-\alpha$ CI for μ
  2. $\bar{X} + t_{\alpha,n-1}\frac{S}{\sqrt{n}}$ is an approximate $1-\alpha$ upper confidence bound for μ
  3. $\bar{X} - t_{\alpha,n-1}\frac{S}{\sqrt{n}}$ is an approximate $1-\alpha$ lower confidence bound for μ.

     When n is large, S should be close to σ, and $t_{\alpha,n-1}$ should be close to z_{α}, thus these two types of approximations should give very similar results

3.2 Confidence Interval for Proportion with Large Sample Size

As we mentioned, when the population is not normal, we can often use CLT to give the approximate CI for the population mean. Here we focus on a special case: constructing confidence intervals on the population proportion.

Let $X_1, \ldots, X_n$ be iid samples from Bernoulli distribution with success probability p. Using CLT we know that the **sample proportion**, $\hat{p} = \bar{X} = \frac{1}{n}\sum X_i$, is approximately normal with mean p and variance $p(1-p)/n$, when p is not too close to either 0 or 1 and if n is relatively large, i.e.,

$$Z = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \sim_{\text{approx}} N(0,1)$$

Typically, to apply this approximation we require that np and $np(1-p)$ be greater than or equal to 5.

Thus we have

$$\mathbf{P}(-z_{\alpha/2} \leq \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \leq z_{\alpha/2}) \approx 1-\alpha$$

This can be rearranged as

$$\mathbf{P}(\hat{p} - z_{\alpha/2}\sqrt{p(1-p)/n} \leq p \leq \hat{p} + z_{\alpha/2}\sqrt{p(1-p)/n}) \approx 1-\alpha$$

Of course we can solve the above two inequalities (quadratic inequalities for p) to derive the CI for p. But in practice, a satisfactory solution is usually obtained directly by replacing the unknown p with $\hat{p}$,

which results in

$$\mathbf{P}(\hat{p} - z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n} \le p \le \hat{p} + z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}) \approx 1 - \alpha$$

this leads to the approximate $1-\alpha$ CI for p as $\hat{p} \pm z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}$.

Similarly, we have the approximate $1-\alpha$ lower confidence bound for p as $\hat{p} - z_{\alpha}\sqrt{\hat{p}(1-\hat{p})/n}$ and the upper confidence bound $\hat{p} + z_{\alpha}\sqrt{\hat{p}(1-\hat{p})/n}$.

Remark: Note that $p \in [0,1]$, thus if the lower limit is negative, we will use 0 instead, and if the upper limit is larger than 1, we will use 1 instead.

Choice of sample size: The estimation error is $E := |\hat{p} - p|$, thus we are approximately $1-\alpha$ confident that $E \le z_{\alpha/2}\sqrt{p(1-p)/n}$. Thus the sample size needed to meet the desired error E is

$$n = \lceil \Big(\frac{z_{\alpha/2}}{E}\Big)^2 p(1-p) \rceil$$

However, in practice when determining the required sample size, an estimate of p is required to use the above equation, thus $\hat{p}$ can be substituted. Or an alternative approach to choosing n uses the fact that $p(1-p) \le 1/4$ and thus an upper bound for the sample size is

$$n = \lceil \Big(\frac{z_{\alpha/2}}{E}\Big)^2 \frac{1}{4} \rceil$$

3.3 Confidence interval using maximum likelihood estimator

We have established the maximum likelihood estimator(MLE), then we are interested in its bias and variance. Asymptotically (i.e., when the sample size is sufficiently large), the bias of MLE is negligible compared to the standard deviation, so we will only care about its variance.

If $\theta \in \mathbf{R}$, define the **score function** as the derivative of the log-likelihood function

$$U(\theta) = \frac{\partial \ell(\theta)}{\partial \theta}.$$

Note that $U(\theta)$ depends on the sample values $x_1, \ldots, x_n$ because the likelihood function $L(\theta) = f(x_1, \ldots, x_n; \theta)$, and thus $U(\theta)$ is also a random variable.

We also note the expectation (taken with respect to $X_1, \ldots, X_n$) of $U(\theta)$ is zero:

$$\mathbf{E}[U(\theta)] = \mathbf{E}[\frac{\partial \log L(\theta)}{\partial \theta}] = \mathbf{E}[\frac{1}{L(\theta)}\frac{\partial L(\theta)}{\partial \theta}] = \frac{\partial}{\partial \theta}(\int \cdots \int L(\theta) dx_1 \ldots dx_n) = \frac{\partial 1}{\partial \theta} = 0.$$

We define the **'Fisher information'** as the variance of $U(\theta)$

$$I(\theta) = \mathbf{E}\left(\frac{\partial \ell(\theta)}{\partial \theta}\right)^2 \tag{5}$$

If $\ell''(\theta)$ exists, then

$$I(\theta) = -\mathbf{E}\frac{\partial^2 \ell(\theta)}{\partial \theta^2}.$$

Then Fisher showed that, if the data distribution is indeed in the parameter group, then

$$\sqrt{I(\theta)}(\widehat{\theta} - \theta) \to N(0, 1),$$

where $\widehat{\theta}$ is the MLE. In particular, the variance of $\widehat{\theta}$ is $1/I(\theta)$ (here we stress that the θ here is the true underlying parameter of the data distribution). Therefore, according to the probability review lecture notes, by selecting $c_{2.5\%}$ and $c_{97.5\%}$ to be the 2.5 and 97.5% quantile of standard normal distribution(a fixed number), we have

$$\lim_{n\to\infty} Prob(c_{2.5\%} \leq \sqrt{I(\theta)}(\widehat{\theta} - \theta) \leq c_{97.5\%}) = 95\%,$$

which forms the theoretical foundation for the construction of confidence intervals.

Example 2 Variance of MLE for binary random variable Suppose the data $x_1, \cdots, x_n$ satisfy binomial distribution, i.e., $\mathbf{P}(x_i = 1) = p$ and $\mathbf{P}(x_i = 0) = 1 - p$, then the likelihood function is

$$L(p) = p^S(1-p)^{n-S} \quad \text{here } S = \sum_{i=1}^{n} x_i$$

and the log-likelihood function is

$$\ell(p) = S\log p + (n - S)\log(1 - p).$$

So we have

$$\frac{\partial \ell(p)}{\partial p} = \frac{S}{p} - \frac{n-S}{1-p}, \text{ and } \mathbf{E}\left(\frac{\partial \ell(p)}{\partial p}\right)^2 = \frac{1}{p^2(1-p)^2} \times \mathbf{E}S^2 + \frac{n^2}{(1-p)^2} - \frac{2n}{p(1-p)^2}\mathbf{E}S$$

Since $\mathbf{E}S^2 = n^2p^2 + np(1-p)$ and $\mathbf{E}S = np$, we have

$$\mathbf{E}\left(\frac{\partial \ell(p)}{\partial p}\right)^2 = \frac{n^2p^2 + np(1-p)}{p^2(1-p)^2} + \frac{n^2}{(1-p)^2} - \frac{2n^2p}{p(1-p)^2} = \frac{n}{p(1-p)}$$

Thus we know that the fisher information $I(p) = \frac{n}{p(1-p)}$. So the variance of the MLE estimator $\widehat{p}$ is $\text{Var}(\widehat{p}) = 1/I(\theta) = \frac{p(1-p)}{n}$. In practice, since p is unknown, we replace p with its estimator $\widehat{p}$, then the estimated variance should be $\frac{\widehat{p}(1-\widehat{p})}{n}$.

Example 3 Variance of normal mean estimator Suppose we know the variance σ^2, the log likelihood

function is

$$\ell(\mu) = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2 - \log((2\pi)^{n/2}\sigma^n)$$
$$\Rightarrow \frac{\partial \ell(\mu)}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu)$$
$$\Rightarrow I(\mu) = \mathbf{E}\left(\frac{\partial \ell(\mu)}{\partial \mu}\right)^2 = \frac{n\sigma^2}{\sigma^4} = \frac{n}{\sigma^2}$$

Thus $I(\mu) = \frac{n}{\sigma^2}$ and the variance $\mathrm{Var}(\widehat{\mu}) = \frac{\sigma^2}{n}$. Note that we may also compute the variance of $\hat{\mu} = \bar{X}$ directly.

Example 4 Variance of normal variance estimator Now suppose the mean is known. Then the log likelihood function is

$$\ell(\sigma^2) = \frac{-1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2 - \log((2\pi)^{n/2}(\sigma^2)^{n/2})$$
$$\Rightarrow \frac{\partial \ell(\sigma^2)}{\partial \sigma^2} = \frac{1}{2\sigma^4}\sum_{i=1}^{n}(x_i-\mu)^2 - \frac{n}{2\sigma^2}$$
$$\Rightarrow I(\sigma^2) = \mathbf{E}\left(\frac{\partial \ell(\sigma^2)}{\partial \sigma^2}\right)^2 = \frac{\sigma^4}{4\sigma^8}\times n(n+2) + \frac{n^2}{4\sigma^4} - \frac{n^2\sigma^2}{2\sigma^6} = \frac{n}{2\sigma^4}.$$

So the variance of the MLE $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2$ is $\frac{2\sigma^4}{n}$.

Theorem 3.1 (**Cramér–Rao lower bound**). *Suppose θ is an unknown deterministic parameter that is to be estimated from n independent observations (measurements) of X, each from a distribution according to some probability density function $f(x;\theta)$. The variance of any unbiased estimator $\hat{\theta}$ of θ is lower bounded by the reciprocal of the Fisher information:*

$$\mathrm{Var}(\hat{\theta}) \geq \frac{1}{I(\theta)}.$$

Proof Suppose $T = T(X_1, X_2, \ldots, X_n)$ is an unbiased estimator of θ. Recall the score function is defined as $U = \frac{\partial \ell(\theta)}{\partial \theta}$. Note that

$$\mathbf{E}[T\cdot U] = \mathbf{E}[T\frac{\partial \log L(\theta)}{\partial \theta}] = \mathbf{E}[T\frac{1}{L(\theta)}\frac{\partial L(\theta)}{\partial \theta}]$$
$$= \int_{-\infty}^{\infty}\cdots\int_{-\infty}^{\infty} T(x_1,\ldots,x_n)\frac{1}{f(x_1;\theta)\cdots f(x_n;\theta)}\frac{\partial f(x_1;\theta)\cdots f(x_n;\theta)}{\partial\theta}\cdot f(x_1;\theta)\cdots f(x_n;\theta)dx_1\cdots dx_n$$
$$= \frac{\partial}{\partial\theta}\mathbf{E}[T] = \frac{\partial\theta}{\partial\theta} = 1.$$

By the Cauchy–Schwarz inequality inequality, we have

$$\sqrt{\mathrm{Var}(T)\mathrm{Var}(U)} \geq |\mathrm{Cov}(T,U)| = |\mathbf{E}(TU) - \mathbf{E}[T]\mathbf{E}[U]| = |\mathbf{E}[TU]| = 1, \quad \text{since } \mathbf{E}[U] = 0,$$

thus we have $\mathrm{Var}(T) \geq \frac{1}{\mathrm{Var}(U)} = \frac{1}{I(\theta)}$, for any unbiased estimator T of θ. □

Suppose the observations $X_1, X_2, \cdots, X_n$ are from a parametric group $F(;\theta)$, and we want to estimate the parameter θ. Suppose the maximum likelihood estimator is $\widehat{\theta}$, recall the following:

Theorem 3.2. *Suppose $X_1, \cdots, X_n$ are sampled from a parametric distribution $F(;\theta)$ and the maximum likelihood estimator for θ is $\widehat{\theta}$, then*

$$\sqrt{I(\theta)}(\widehat{\theta} - \theta) \to_d N(0,1) \tag{6}$$

Above, the Fisher information

$$\begin{aligned} I_n(\theta) &= \mathbf{E}\left(\frac{\partial \ell(\theta; X_1, \ldots, X_n)}{\partial \theta}\right)^2 = \mathbf{E}\left(\frac{\partial [\ell(\theta; X_1) + \ell(\theta; X_2) + \cdots + \ell(\theta; X_n)]}{\partial \theta}\right)^2 \\ &= \sum_{i=1}^{n} \mathbf{E}\left(\frac{\partial \ell(\theta; X_1)}{\partial \theta}\right)^2 = nI_1(\theta), \end{aligned}$$

where $I_1(\theta)$ denotes the Fisher information in one sample. Therefore, (6) translates into

$$\sqrt{n}\sqrt{I_1(\theta)}(\widehat{\theta} - \theta) \to_d N(0,1).$$

In order to construct the (two-sided) confidence interval, we first need a consistent estimator for the Fisher information, i.e., find an estimator $\widehat{I}_1(\theta)$ such that

$$\begin{aligned} \widehat{I}_1(\theta) &\to_p I_1(\theta) \\ \text{then } \sqrt{n \times \widehat{I}_1(\theta)} \times (\widehat{\theta} - \theta) &\to_d N(0,1) \end{aligned}$$

Thus $\sqrt{n \times \widehat{I}_1(\theta)} \times (\widehat{\theta} - \theta)$ is a pivot.

To establish the confidence interval, suppose $z_{\alpha/2}$ still being the upper $\alpha/2$ quantile of a standard normal random variable, define the confidence interval

$$\left[\widehat{\theta} - \frac{z_{\frac{\alpha}{2}}}{\sqrt{n\widehat{I}_1(\theta)}},\ \ \widehat{\theta} + \frac{z_{\frac{\alpha}{2}}}{\sqrt{n\widehat{I}_1(\theta)}}\right]$$

then we have

$$\mathbf{P}\left(\widehat{\theta} - \frac{z_{\frac{\alpha}{2}}}{\sqrt{n\widehat{I}_1(\theta)}} \leq \theta \leq \widehat{\theta} + \frac{z_{\frac{\alpha}{2}}}{\sqrt{n\widehat{I}_1(\theta)}}\right) \to 1 - \alpha \tag{7}$$

The one-side confidence interval has a similar situation. Suppose z_α being the upper α quantile of a

standard normal random variable, consider the interval $(-\infty, \widehat{\theta} + \frac{z_\alpha}{\sqrt{n\widehat{I}_1(\theta)}}]$, then

$$\mathbf{P}\left(\theta \in (-\infty, \widehat{\theta} + \frac{z_\alpha}{\sqrt{n\widehat{I}_1(\theta)}})\right) = \mathbf{P}\left(\sqrt{n\widehat{I}_1(\theta)} \times (\widehat{\theta} - \theta) \geq -z_\alpha\right) = 1 - \alpha$$

For example, for normal random variable with unknown mean μ and unknown variance σ^2, the fisher information for μ is $I_1(\mu) = 1/\sigma^2$ and $I_n(\theta) = n/\sigma^2$, since the the MLE $\hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$ (or the sample variance S^2) are consistent estimators of σ^2, and the MLE for μ is $\hat{\mu} = \bar{X}$, we can obtain the confidence interval for μ as

$$\left[\bar{X} - \frac{z_{\frac{\alpha}{2}}}{\sqrt{n/\hat{\sigma}^2}},\ \ \bar{X} + \frac{z_{\frac{\alpha}{2}}}{\sqrt{n/\hat{\sigma}^2}}\right],$$

which is the same as the confidence interval obtained in Section 2. We remark that such method for obtaining the confidence interval through MLE is widely applicable, and can be applied if we cannot find other pivot quantities easily.

Example 5 Confidence interval for exponential distribution Suppose $Y_1, \cdots, Y_n$ satisfy exponential distribution, i.e., Y_i's density is

$$f(x, \lambda) = \lambda \exp(-\lambda x)$$

In order to estimate the parameter λ, we can use the maximum likelihood estimation, i.e., the log $-$likelihood function is given by

$$L(\lambda) = -\lambda(\sum_{i=1}^{n} Y_i) + n\log(\lambda)$$
$$\Rightarrow L'(\lambda) = \frac{n}{\lambda} - \sum_{i=1}^{n} Y_i \text{ and } L''(\lambda) = -n\lambda^{-2}$$

Setting $L' = 0$, we have $\widehat{\lambda} = \frac{n}{\sum_{i=1}^{n} Y_i} = 1/\bar{Y}$. Besides,

$$I(\theta) = \mathbf{E}\left(L'(\lambda)\right)^2 = \frac{n}{\lambda^2} \text{ and from the property of MLE } \frac{\sqrt{n}}{\lambda} \times (\widehat{\lambda} - \lambda) \to_d N(0, 1)$$

For $\mathrm{Var}(\widehat{\lambda}) = \frac{\lambda^2}{n}$, we have $\widehat{\lambda} \to_p \lambda$, so from Slutsky's theorem

$$\frac{\sqrt{n}}{\widehat{\lambda}} \times (\widehat{\lambda} - \lambda) = \frac{\sqrt{n}}{\lambda} \times (\widehat{\lambda} - \lambda) \times \frac{\lambda}{\widehat{\lambda}} \to_d N(0, 1)$$

Choose $z_{\alpha/2}$ the upper $\alpha/2$ quantile of standard normal random variable, then we have

$$\widehat{\lambda} - \frac{\widehat{\lambda}}{\sqrt{n}} z_{\alpha/2} \leq \lambda \leq \widehat{\lambda} + \frac{\widehat{\lambda}}{\sqrt{n}} z_{\alpha/2} \tag{8}$$

in other words, the interval $[\widehat{\lambda} - \frac{\widehat{\lambda}}{\sqrt{n}} z_{\alpha/2}, \widehat{\lambda} + \frac{\widehat{\lambda}}{\sqrt{n}} z_{\alpha/2}]$ is the $1 - \alpha$ confidence interval for λ.

4 CI for Two Samples

4.1 CI for Difference in Means

4.1.1 CI for Difference in Means, Variances Known

Let $X_{11}, \ldots, X_{1n_1}$ be iid samples from $N(\mu_1, \sigma_1^2)$, and $X_{21}, \ldots, X_{2n_2}$ be iid samples from $N(\mu_2, \sigma_2^2)$. Assume σ_1^2, σ_2^2 are known. Denote the sample mean

$$\bar{X}_1 = \frac{1}{n_1}\sum_{i=1}^{n_1} X_{1i}, \quad \bar{X}_2 = \frac{1}{n_2}\sum_{i=1}^{n_2} X_{2i}.$$

Then we have

$$\frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})}} \sim N(0,1)$$

Thus the CI on the difference in two means $\mu_1 - \mu_2$ is

1. The two sided CI is

$$[\bar{X}_1 - \bar{X}_2 - z_{\alpha/2}\sqrt{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})},\ \bar{X}_1 - \bar{X}_2 + z_{\alpha/2}\sqrt{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})}]$$

2. The one-sided $1-\alpha$ upper confidence bound is $\bar{X}_1 - \bar{X}_2 + z_\alpha\sqrt{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})}$

3. The one-sided $1-\alpha$ lower confidence bound is $\bar{X}_1 - \bar{X}_2 - z_\alpha\sqrt{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})}$

Similarly, when the data distribution is not normal, using the CLT, we get the approximate CI using the above formulas.

If we further assume equal sample size $n_1 = n_2 = n$, similar to the one-sample case, we can derive the sample size needed for the desired estimation error $E = |(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)|$ is:

$$n = \lceil \Big(\frac{z_{\alpha/2}}{E}\Big)^2 (\sigma_1^2 + \sigma_2^2) \rceil$$

4.1.2 CI for Difference in Means, Variances unknown but equal

Let $X_{11}, \ldots, X_{1n_1}$ be iid samples from $N(\mu_1, \sigma_1^2)$, and $X_{21}, \ldots, X_{2n_2}$ be iid samples from $N(\mu_2, \sigma_2^2)$. Assume σ_1^2, σ_2^2 are unknown but equal $\sigma_1^2 = \sigma_2^2 = \sigma^2$.

We know that

$$\frac{(n_1-1)S_1^2}{\sigma^2} \sim \chi^2(n_1 - 1), \quad \frac{(n_2-1)S_2^2}{\sigma^2} \sim \chi^2(n_2 - 1)$$

and they are independent. Thus we have

$$\frac{(n_1-1)S_1^2}{\sigma^2} + \frac{(n_2-1)S_2^2}{\sigma^2} \sim \chi^2(n_1 + n_2 - 2)$$

We define the so-called **pooled estimator** of σ^2:

$$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}.$$

Meanwhile we have

$$\frac{\bar{X}_1 - \bar{X}_2 - (\mu_1-\mu_2)}{\sqrt{(\frac{\sigma^2}{n_1}+\frac{\sigma^2}{n_2})}} \sim N(0,1)$$

By the definition of t distribution we thus have

$$\frac{\bar{X}_1 - \bar{X}_2 - (\mu_1-\mu_2)}{S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \sim t(n_1+n_2-2).$$

Thus we have the CI for $\mu_1 - \mu_2$ is

1. The two sided $1-\alpha$ CI is

$$[\bar{X}_1 - \bar{X}_2 - t_{\alpha/2,n_1+n_2-2}S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}},\ \bar{X}_1 - \bar{X}_2 + t_{\alpha/2,n_1+n_2-2}S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}]$$

2. The one-sided $1-\alpha$ upper confidence bound is $\bar{X}_1 - \bar{X}_2 + t_{\alpha,n_1+n_2-2}S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}$

3. The one-sided $1-\alpha$ lower confidence bound is $\bar{X}_1 - \bar{X}_2 - t_{\alpha,n_1+n_2-2}S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}$

4.2 Confidence interval for variance ratio

Let $X_{11},\ldots,X_{1n_1}$ be iid samples from $N(\mu_1,\sigma_1^2)$, and $X_{21},\ldots,X_{2n_2}$ be iid samples from $N(\mu_2,\sigma_2^2)$. We know that

$$\frac{(n_1-1)S_1^2}{\sigma_1^2} \sim \chi^2(n_1-1), \quad \frac{(n_2-1)S_2^2}{\sigma_2^2} \sim \chi^2(n_2-1)$$

and they are independent. By the definition of F distribution, we have

$$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F_{n_1-1,n_2-1}$$

Thus

1. The two-sided $(1-\alpha)$ CI for σ_1^2/σ_2^2 is

$$\frac{S_1^2}{S_2^2}\frac{1}{F_{\alpha/2}(n_1-1,n_2-1)} \le \frac{\sigma_1^2}{\sigma_2^2} \le \frac{S_1^2}{S_2^2}\frac{1}{F_{1-\alpha/2}(n_1-1,n_2-1)}$$

   which is

$$\frac{S_1^2}{S_2^2}F_{1-\alpha/2}(n_2-1,n_1-1) \le \frac{\sigma_1^2}{\sigma_2^2} \le \frac{S_1^2}{S_2^2}F_{\alpha/2}(n_2-1,n_1-1)$$

2. The one-sided $1-\alpha$ upper confidence bound for σ_1^2/σ_2^2 is $\frac{S_1^2}{S_2^2}F_\alpha(n_2-1,n_1-1)$

3. The one-sided $1-\alpha$ lower confidence bound is $\frac{S_1^2}{S_2^2}F_{1-\alpha}(n_2-1,n_1-1)$

4.3 CI for Difference in Proportions, large sample size

Let $X_{11},\ldots,X_{1n_1}$ be iid samples from Bernoulli distribution with success probability p_1, and $X_{21},\ldots,X_{2n_2}$ be iid samples from Bernoulli distribution with success probability p_2.

We first deal with a special case: suppose $n_1=n_2=n$. In that case we have

$$\widehat{p}_1-\widehat{p}_2=\frac{1}{n}\sum_{i=1}^{n}X_{1i}-\frac{1}{n}\sum_{i=1}^{n}X_{2i}=\frac{1}{n}\sum_{i=1}^{n}(X_{1i}-X_{2i}) \tag{9}$$

For $X_{1i}-X_{2i}, i=1,\cdots,n$ are i.i.d., with mean p_1-p_2 and variance $p_1(1-p_1)+p_2(1-p_2)$, we have

$$\frac{\sqrt{n}(\widehat{p}_1-\widehat{p}_2-p_1+p_2)}{\sqrt{p_1(1-p_1)+p_2(1-p_2)}}\to_d N(0,1) \tag{10}$$

For $\widehat{p}_1\to p_1$, $\widehat{p}_2\to_p p_2$, we have

$$\widehat{p}_1(1-\widehat{p}_1)+\widehat{p}_2(1-\widehat{p}_2)\to_p p_1(1-p_1)+p_2(1-p_2)$$

and therefore from Slutsky's theorem

$$\begin{aligned}\frac{\sqrt{n}(\widehat{p}_1-\widehat{p}_2-p_1+p_2)}{\sqrt{\widehat{p}_1(1-\widehat{p}_1)+\widehat{p}_2(1-\widehat{p}_2)}}&=\frac{\sqrt{n}(\widehat{p}_1-\widehat{p}_2-p_1+p_2)}{\sqrt{p_1(1-p_1)+p_2(1-p_2)}}\times\frac{\sqrt{p_1(1-p_1)+p_2(1-p_2)}}{\sqrt{\widehat{p}_1(1-\widehat{p}_1)+\widehat{p}_2(1-\widehat{p}_2)}}\\&\to_d N(0,1)\end{aligned}$$

In particular, we have

$$\begin{aligned}&P\left(\widehat{p}_1-\widehat{p}_2-\frac{\sqrt{\widehat{p}_1(1-\widehat{p}_1)+\widehat{p}_2(1-\widehat{p}_2)}}{\sqrt{n}}\times z_{\alpha/2}\le p_1-p_2\le\widehat{p}_1-\widehat{p}_2+\frac{\sqrt{\widehat{p}_1(1-\widehat{p}_1)+\widehat{p}_2(1-\widehat{p}_2)}}{\sqrt{n}}\times z_{\alpha/2}\right)\\&\to 1-\alpha\end{aligned}$$

and we derive the two-side $1-\alpha$ confidence interval for p_1-p_2.

Therefore, choose $z_{\alpha/2}$ as the upper $\alpha/2$ quantile of standard normal random variable, we have

$$\lim_{n\to\infty}P\left(\frac{\sqrt{n}(\widehat{p}_1-\widehat{p}_2-p_1+p_2)}{\sqrt{\widehat{p}_1(1-\widehat{p}_1)+\widehat{p}_2(1-\widehat{p}_2)}}\in[-z_{\alpha/2},z_{\alpha/2}]\right)=1-\alpha$$

Denote the sample proportion, $\hat{p}_1=\bar{X}_1=\frac{1}{n_1}\sum X_{1i}$, $\hat{p}_2=\bar{X}_2=\frac{1}{n_2}\sum X_{2i}$, are approximately normal

and thus

$$Z = \frac{\hat{p}_1 - \hat{p}_2 - (p_1 - p_2)}{\sqrt{(p_1(1-p_1)/n_1 + (p_2(1-p_2)/n_2)}} \sim_{\text{approx}} N(0,1)$$

And the approximate CI for $p_1 - p_2$ is

1. The approximate two sided $(1-\alpha)$ CI for $p_1 - p_2$ is

$$[\hat{p}_1 - \hat{p}_2 - z_{\alpha/2}\sqrt{(\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2})},\ \hat{p}_1 - \hat{p}_2 + z_{\alpha/2}\sqrt{(\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2})}]$$

2. The approximate one-sided $1-\alpha$ upper confidence bound is $\hat{p}_1 - \hat{p}_2 + z_\alpha\sqrt{(\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2})}$

3. The one-sided $1-\alpha$ lower confidence bound is $\hat{p}_1 - \hat{p}_2 - z_\alpha\sqrt{(\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2})}$

5 Summary

After this lecture, you should realize that:

i. As you can see, a complete statistical inference not only contains the point-estimation, but also should incorporate the information in variance/distribution.

ii. The key for interval estimation is to establish a pivot, e.g., via studentization.

		One Population	Two Populations
Sample Mean	Known Variance	$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$	$\frac{\bar{X}_1-\bar{X}_2-(\mu_1-\mu_2)}{\sqrt{(\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2})}} \sim N(0,1)$
	Unknown (equal) Variance	$\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t(n-1)$	$\frac{\bar{X}_1-\bar{X}_2-(\mu_1-\mu_2)}{S_p\sqrt{(\frac{1}{n_1}+\frac{1}{n_2})}} \sim t(n_1+n_2-2)$
Sample Proportion		$\frac{\hat{p}-p}{\sqrt{p(1-p)/n}} \dot{\sim} N(0,1)$	$\frac{\hat{p}_1-\hat{p}_2-(p_1-p_2)}{\sqrt{(p_1(1-p_1)/n_1+(p_2(1-p_2)/n_2)}} \dot{\sim} N(0,1)$
Sample Variance		$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$	$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F_{n_1-1,n_2-1}$

Table 1: Sampling Distributions

References

Hogg, Robert V., Elliot A. Tanis, and Dale L. Zimmerman (2018). *Probability and statistical inference.*

		One Population	Two Populations
Sample Mean	Known Variance	$[\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}]$ $[\bar{X} - z_{\alpha}\frac{\sigma}{\sqrt{n}},\ \infty)$, $(-\infty,\ \bar{X} + z_{\alpha}\frac{\sigma}{\sqrt{n}}]$	$\bar{X}_1 - \bar{X}_2 \pm z_{\alpha/2}\sqrt{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})}$ $[\bar{X}_1 - \bar{X}_2 - z_{\alpha}\sqrt{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})},\ \infty)$ $(-\infty,\ \bar{X}_1 - \bar{X}_2 + z_{\alpha}\sqrt{(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})}]$
	Unknown (equal) Variance	$[\bar{X} - t_{\alpha/2,n-1}\frac{S}{\sqrt{n}},\ \bar{X} + t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}]$ $[\bar{X} - t_{\alpha,n-1}\frac{S}{\sqrt{n}},\ \infty)$ $(-\infty,\ \bar{X} + t_{\alpha,n-1}\frac{S}{\sqrt{n}}]$	$\bar{X}_1 - \bar{X}_2 \pm t_{\alpha/2,n_1+n_2-2}S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$ $[\bar{X}_1 - \bar{X}_2 - t_{\alpha,n_1+n_2-2}S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}},\ \infty)$ $(-\infty,\ \bar{X}_1 - \bar{X}_2 + t_{\alpha,n_1+n_2-2}S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}]$
Sample Proportion		$\hat{p} \pm z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}$ $[\hat{p} - z_{\alpha}\sqrt{\hat{p}(1-\hat{p})/n},\ \infty)$ $(-\infty,\ \hat{p} + z_{\alpha}\sqrt{\hat{p}(1-\hat{p})/n}]$	$\hat{p}_1 - \hat{p}_2 \pm z_{\alpha/2}\sqrt{(\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2})}$ $[\hat{p}_1 - \hat{p}_2 - z_{\alpha}\sqrt{(\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2})},\ \infty)$ $(-\infty,\ \hat{p}_1 - \hat{p}_2 + z_{\alpha}\sqrt{(\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2})}]$
Sample Variance		$\frac{(n-1)S^2}{\chi^2_{\alpha/2}(n-1)} \le \sigma^2 \le \frac{(n-1)S^2}{\chi^2_{1-\alpha/2}(n-1)}$ $[\frac{(n-1)S^2}{\chi^2_{\alpha}(n-1)},\ \infty)$, $(0,\ \frac{(n-1)S^2}{\chi^2_{1-\alpha}(n-1)}]$	$[\frac{S_1^2}{S^2}F_{1-\alpha/2}(n_2-1, n_1-1),\ \frac{S_1^2}{S^2}F_{\alpha/2}(n_2-1, n_1-1)]$ $[\frac{S_1^2}{S^2}F_{1-\alpha}(n_2-1, n_1-1),\ \infty)$ $(0,\ \frac{S_1^2}{S^2}F_{\alpha}(n_2-1, n_1-1)]$
MLE		$\left[\widehat{\theta} - \frac{z_{\frac{\alpha}{2}}}{\sqrt{n\widehat{I}_1(\theta)}},\ \widehat{\theta} + \frac{z_{\frac{\alpha}{2}}}{\sqrt{n\widehat{I}_1(\theta)}}\right]$ $[\widehat{\theta} - \frac{z_{\alpha}}{\sqrt{n\widehat{I}_1(\theta)}}, +\infty), (-\infty, \widehat{\theta} + \frac{z_{\alpha}}{\sqrt{n\widehat{I}_1(\theta)}}]$	

Table 2: Confidence Intervals